



Agile Development Process and Devops Lab (ISWE406P)

LAB FAT

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Seat No: 50

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Slot: L47+L48

Question:

ONLINE TICKET CANCELLATION SYSTEM

Problem Statement:

Ensure tickets are cancelled and refunded correctly.

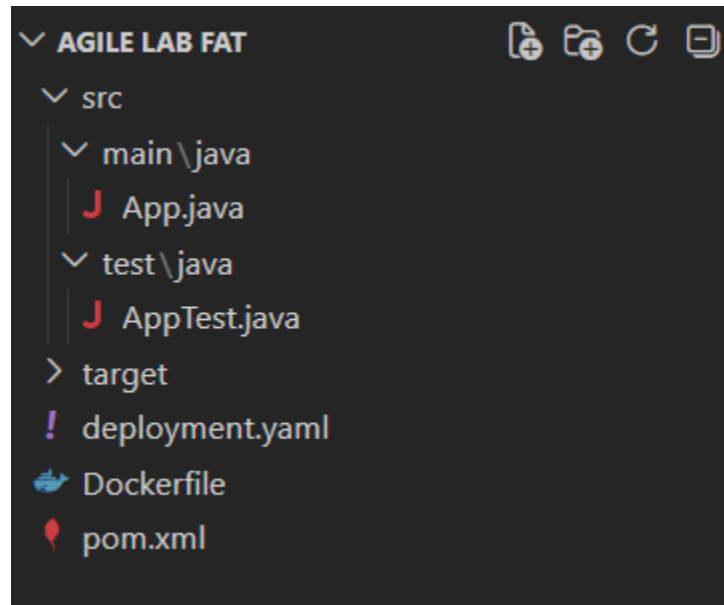
Tasks:

- 1) Develop cancellation and refund logic
- 2) Write JUnit tests for refund scenarios
- 3) Implement CI/CD pipeline with automated testing
- 4) Deploy services on Docker and Kubernetes

Considerations while writing Java & JUnit code:

- For Cancelling Ticket either Refund should be activated or payment should be unsuccessful
- For Refund Activation, the payment status has to be successful and ticket status can be either "Pending" or "Booked"
- If Ticket status is "Cancelled" and if someone has applied for Refund, it is considered as "Invalid Refund Request"

Created Folder Structure:



App.java:

```
src > main > java > App.java
1 public class App {
2
3     public static String ValidateCancellation(boolean refundStatus, String paymentStatus) {
4         if ((refundStatus == true) || paymentStatus.equals("Not successful")) {
5             return "Ticket Cancelled Successfully";
6         } else {
7             return "Ticket not cancelled, please activate refund!";
8         }
9     }
10
11     public static String ValidateRefund(String ticketStatus, String paymentStatus) {
12         if (ticketStatus.equals("Pending") && paymentStatus.equals("Successful")) {
13             return "Refund Accepted";
14         }
15
16         if (ticketStatus.equals("Booked") && paymentStatus.equals("Successful")) {
17             return "Refund Accepted";
18         }
19
20         if (ticketStatus.equals("Cancelled")) {
21             return "Invalid Refund Request";
22         }
23
24         if (paymentStatus.equals("Not successful")) {
25             return "Refund Rejected";
26         }
27
28         return null;
29     }
30 }
```

The screenshot also shows the Windows taskbar at the bottom with the following details:

- Search bar: Search
- Taskbar icons: File Explorer, Visual Studio Code, Docker Desktop, and others.
- Status bar: ENG IN, 06:41 PM, 15-04-2026

```
src > test > java > J AppTest.java
1  import static org.junit.jupiter.api.Assertions.assertEquals;
2  import org.junit.jupiter.api.Test;
3
4  public class AppTest {
5
6      @Test
7      void testValidateRefund() {
8          assertEquals("Refund Accepted", App.ValidateRefund("Pending", "Successful"));
9
10         assertEquals("Refund Rejected", App.ValidateRefund("Pending", "Not successful"));
11
12         assertEquals("Refund Accepted", App.ValidateRefund("Booked", "Successful"));
13
14         assertEquals("Refund Rejected", App.ValidateRefund("Booked", "Not successful"));
15
16         assertEquals("Invalid Refund Request", App.ValidateRefund("Cancelled", "Successful"));
17
18         assertEquals("Invalid Refund Request", App.ValidateRefund("Cancelled", "Not successful"));
19     }
20 }
21
22
```

Pom.xml:

```
pom.xml
1
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
3  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5  http://maven.apache.org/xsd/maven-4.0.0.xsd">
6      <modelVersion>4.0.0</modelVersion>
7      <groupId>com.demo</groupId>
8      <artifactId>test-java-app</artifactId>
9      <version>1.0</version>
10     <packaging>jar</packaging>
11     <name>test-java-app</name>
12     <properties>
13         <maven.compiler.source>17</maven.compiler.source>
14         <maven.compiler.target>17</maven.compiler.target>
15         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
16         <junit.version>5.10.1</junit.version>
17     </properties>
18     <dependencies>
19         <dependency>
20             <groupId>org.junit.jupiter</groupId>
21             <artifactId>junit-jupiter</artifactId>
22             <version>${junit.version}</version>
23             <scope>test</scope>
24         </dependency>
25     </dependencies>
26     <build>
27         <plugins>
28             <plugin>
29                 <groupId>org.apache.maven.plugins</groupId>
30                 <artifactId>maven-compiler-plugin</artifactId>
```

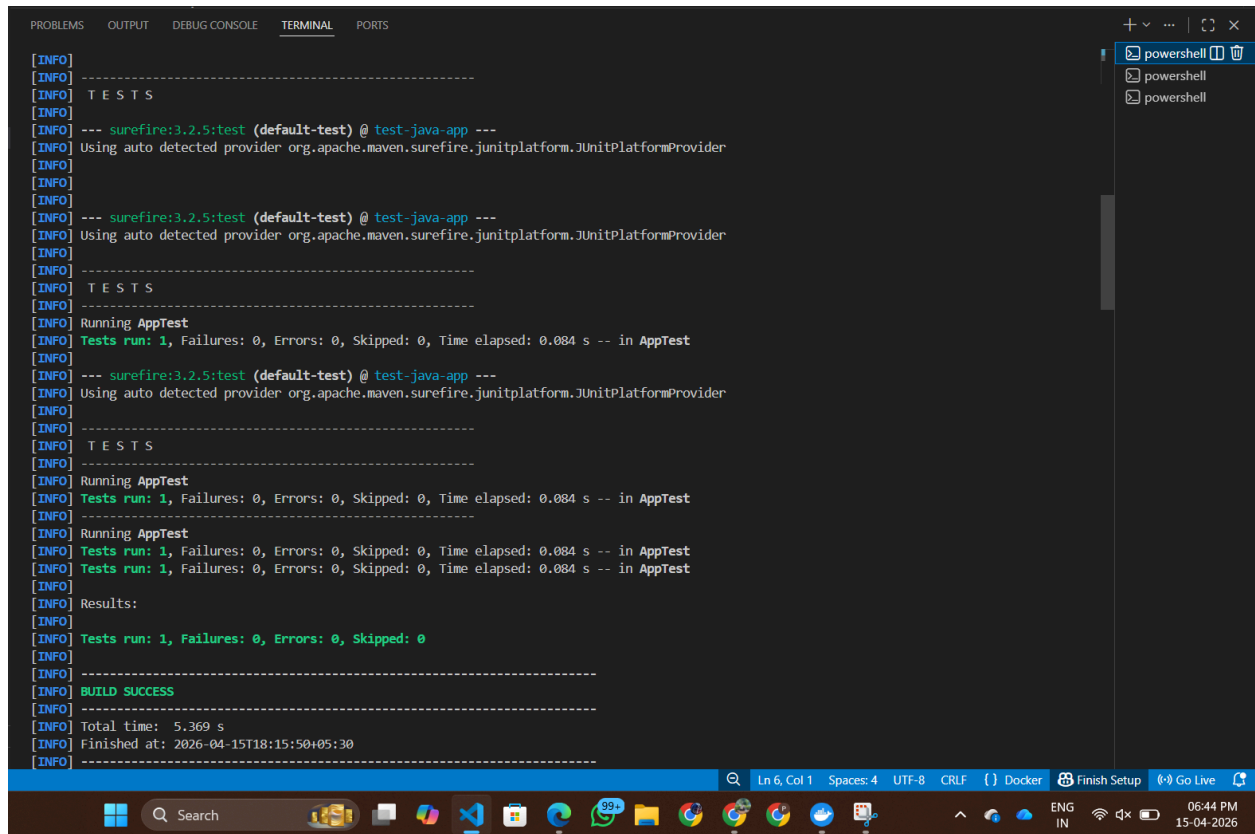
Deployment.yaml:

```
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: simple-java-app
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: simple-java-app
10   template:
11     metadata:
12       labels:
13         app: simple-java-app
14     spec:
15       containers:
16       - name: simple-java-app
17         image: 23mis0563/java-test-app:latest
18         ports:
19         - containerPort: 8080
20
21 ---
22 apiVersion: v1
23 kind: Service
24 metadata:
25   name: java-service
26 spec:
27   type: NodePort
28   selector:
29     app: simple-java-app
30   ports:
```

Dockerfile:

```
Dockerfile
1  FROM eclipse-temurin:17-jdk-alpine
2  WORKDIR /app
3  COPY target/*.jar app.jar
4  EXPOSE 8080
5  ENTRYPOINT ["java", "-jar", "app.jar"]
6
```

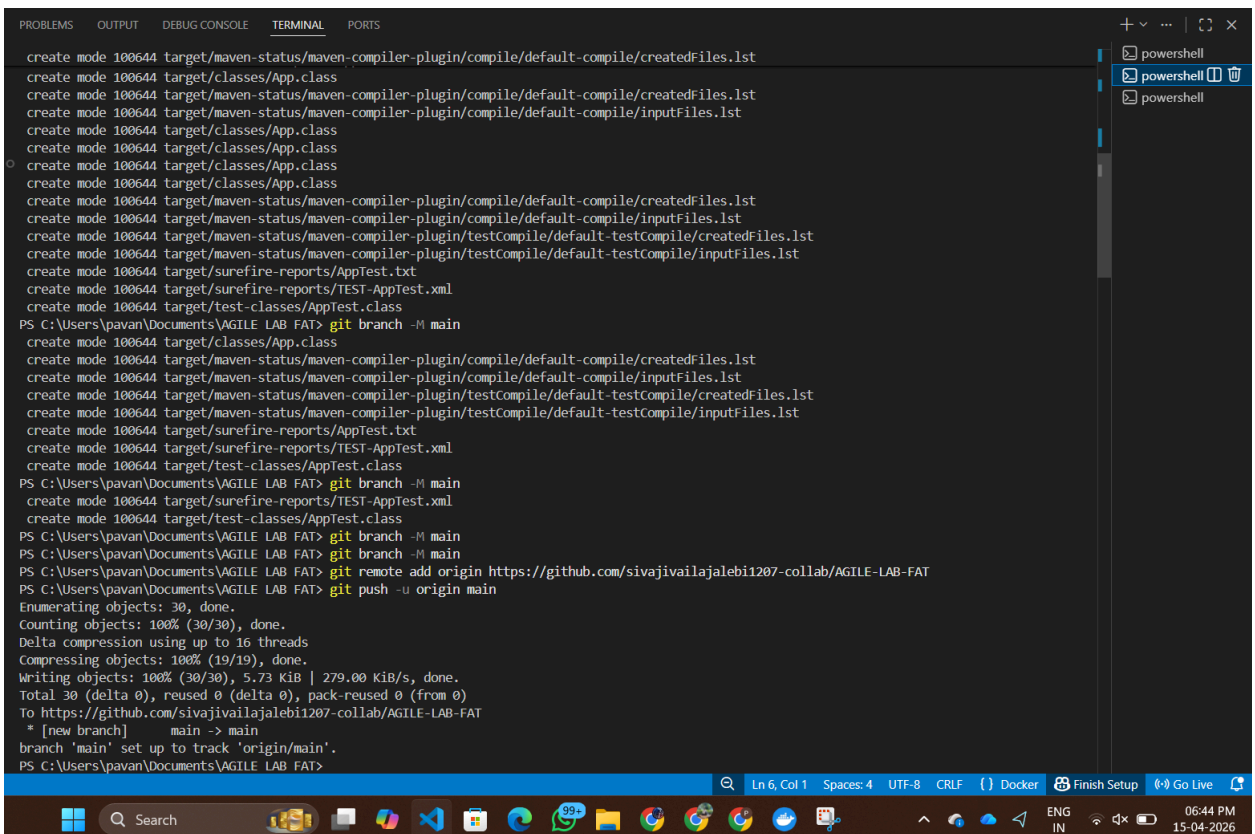
Locally testing:



The screenshot shows an IDE terminal window with the 'TERMINAL' tab selected. The output displays the results of a Maven test execution. It starts with 'INFO' messages, followed by 'TESTS' and 'Running AppTest'. The test results show 'Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest'. This sequence is repeated three times. Finally, it shows 'Results: Tests run: 1, Failures: 0, Errors: 0, Skipped: 0' and 'BUILD SUCCESS'. The total time is 5.369 s, and it finished at 2026-04-15T18:15:50+05:30. The IDE interface includes a taskbar at the bottom with various application icons and a system tray showing the time as 06:44 PM on 15-04-2026.

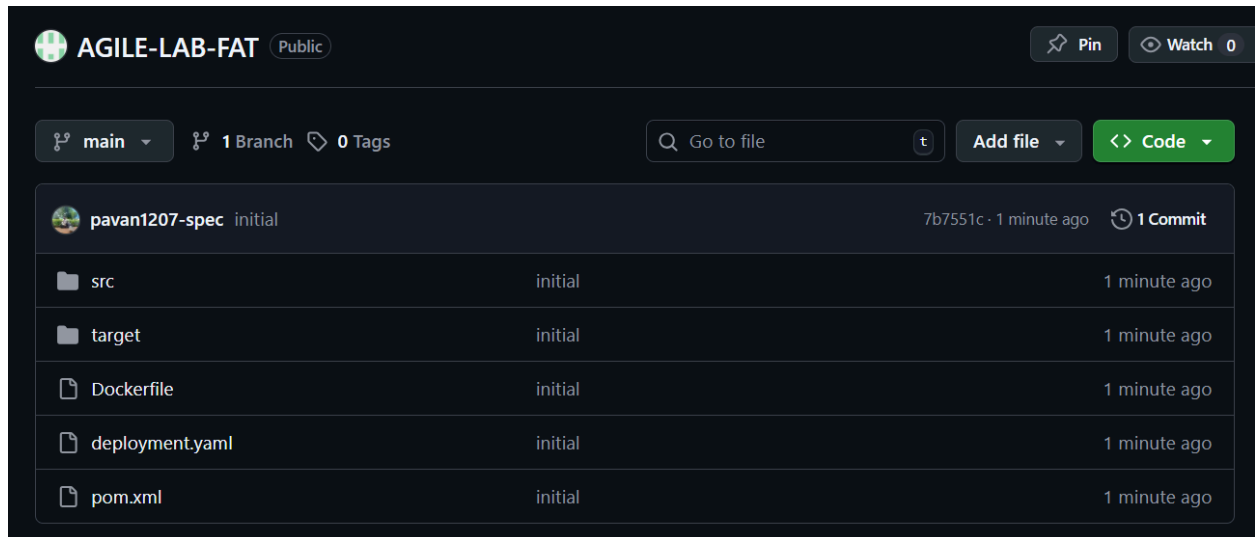
```
[INFO] -----
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] -----
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] -----
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.369 s
[INFO] Finished at: 2026-04-15T18:15:50+05:30
[INFO] -----
```

Pushed this directory into github:

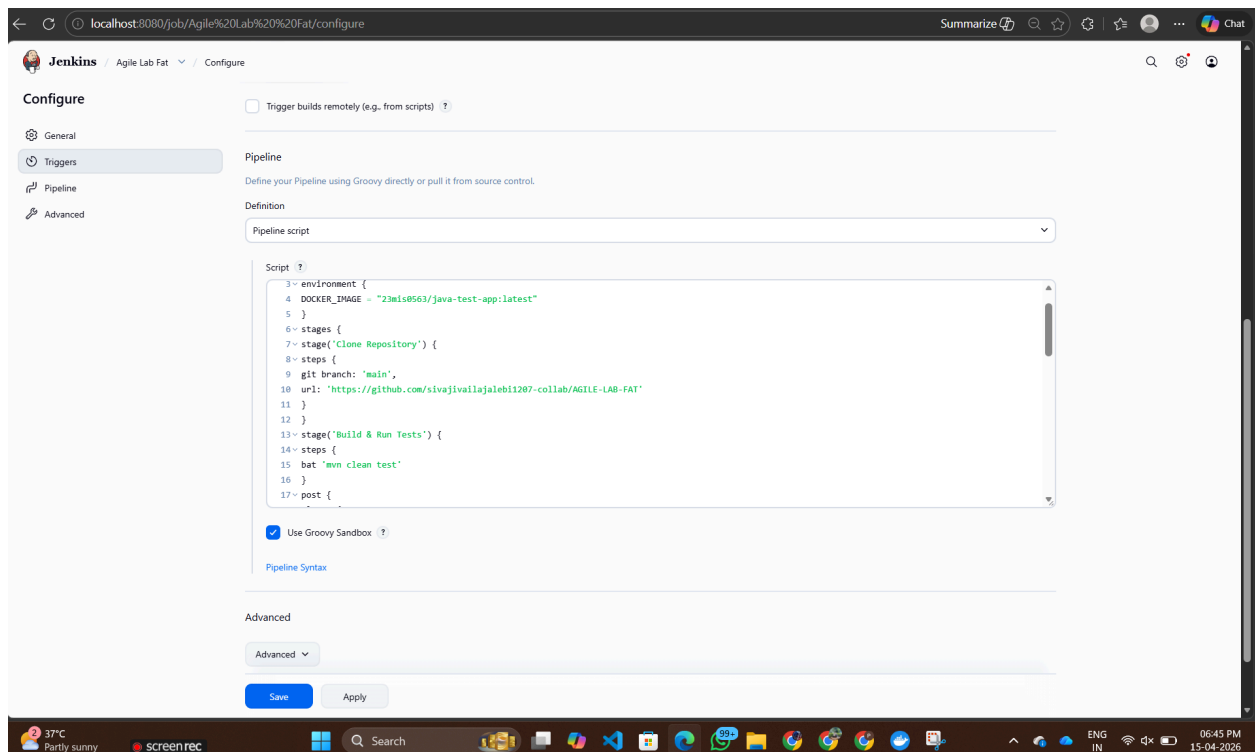


The screenshot shows an IDE terminal window with the 'TERMINAL' tab selected. The output displays a series of git commands and their results. It starts with 'create mode' messages for various files. Then, it shows 'PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main'. This is followed by another 'create mode' message. Then, it shows 'PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main', 'PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main', 'PS C:\Users\pavan\Documents\AGILE LAB FAT> git remote add origin https://github.com/sivajivallajalebi1207-collab/AGILE-LAB-FAT', and 'PS C:\Users\pavan\Documents\AGILE LAB FAT> git push -u origin main'. The output then shows the progress of pushing the code to GitHub, including 'Enumerating objects: 30, done.', 'Counting objects: 100% (30/30), done.', 'Delta compression using up to 16 threads', 'Compressing objects: 100% (19/19), done.', 'Writing objects: 100% (30/30), 5.73 KiB | 279.00 KiB/s, done.', 'Total 30 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)', and 'To https://github.com/sivajivallajalebi1207-collab/AGILE-LAB-FAT'. The final output shows 'branch 'main' set up to track 'origin/main'.' and 'PS C:\Users\pavan\Documents\AGILE LAB FAT>'. The IDE interface includes a taskbar at the bottom with various application icons and a system tray showing the time as 06:44 PM on 15-04-2026.

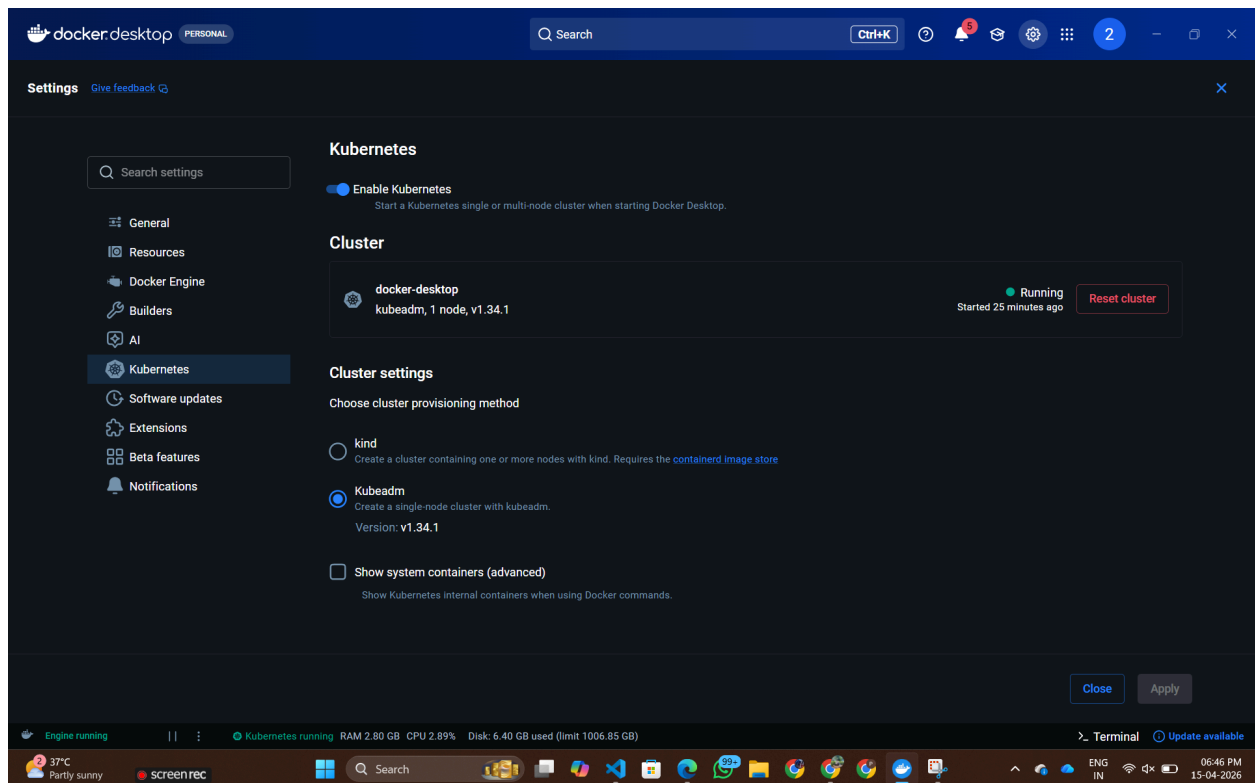
```
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/classes/App.class
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/classes/App.class
create mode 100644 target/classes/App.class
create mode 100644 target/classes/App.class
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/AppTest.txt
create mode 100644 target/surefire-reports/TEST-AppTest.xml
create mode 100644 target/test-classes/AppTest.class
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
create mode 100644 target/classes/App.class
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/AppTest.txt
create mode 100644 target/surefire-reports/TEST-AppTest.xml
create mode 100644 target/test-classes/AppTest.class
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
PS C:\Users\pavan\Documents\AGILE LAB FAT> git remote add origin https://github.com/sivajivallajalebi1207-collab/AGILE-LAB-FAT
PS C:\Users\pavan\Documents\AGILE LAB FAT> git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (30/30), 5.73 KiB | 279.00 KiB/s, done.
Total 30 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sivajivallajalebi1207-collab/AGILE-LAB-FAT
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\pavan\Documents\AGILE LAB FAT>
```



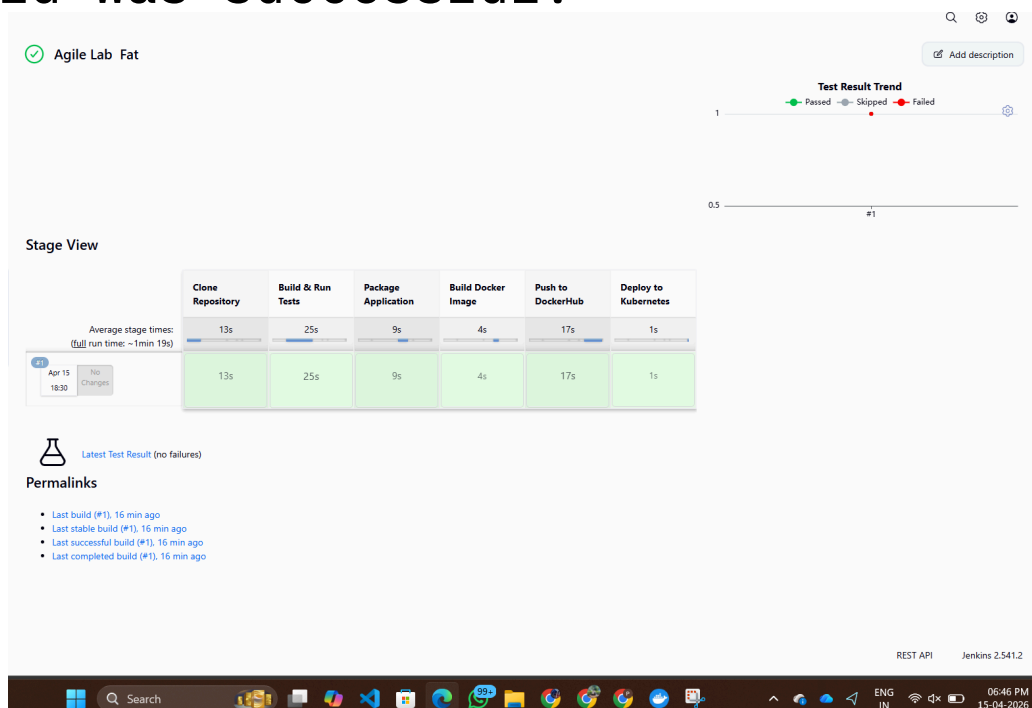
Created a new Pipeline Item in Jenkins and added a pipeline script:



Enabled Kubernetes on Docker Desktop:



Created a new Build in Jenkins and the Build was successful:



Console Output in Jenkins:

```
Console
Download Copy View as plain text

Started by user Pavan Kalyan K
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
> git.exe init C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat # timeout=10
Fetching upstream changes from https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
> git.exe --version # timeout=10
> git --version # 'git version 2.47.1.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 7b7551c47a863a9d6c8c3ac8a43de678fe3a0801 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 7b7551c47a863a9d6c8c3ac8a43de678fe3a0801 # timeout=10
> git.exe branch -a -v --no-abbrev # timeout=10
> git.exe checkout -b main 7b7551c47a863a9d6c8c3ac8a43de678fe3a0801 # timeout=10
Commit message: "initial"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build & Run Tests)
[Pipeline] bat
C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat>mvn clean test
[INFO] Scanning for projects...
```

→ Junit Test by Maven:

```
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.101 s -- in AppTest
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.354 s
[INFO] Finished at: 2026-04-15T18:31:20+05:30
[INFO] -----
Post stage
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Package Application)
[Pipeline] bat
```

→ Pipeline Executed Successfully:

```
82c540078c28: Layer already exists
946c51efe0fd: Layer already exists
13b9be02f27d: Waiting
13b9be02f27d: Waiting
943851fc7b79: Pushed
13b9be02f27d: Pushed
latest: digest: sha256:6403f9eb0a776856267daaed32fc9625c86533e4e37f0c8622136b6fcb20457 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat>set KUBECONFIG=C:/ProgramData/Jenkins/.jenkins/workspace/new_practice/.kube/config

C:\ProgramData\Jenkins\jenkins\workspace\Agile Lab Fat>kubectl apply -f deployment.yaml
deployment.apps/simple-java-app unchanged
service/java-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.2

Deployed and Image Created on Docker Desktop:

